1 Linear Algebra

1.1 What is a vector space?

A vector space V over a field (e.g. R) is a set of vectors that can be added and scaled while staying in V. Think of it as a mathematical "playground" closed under linear operations.

$$\text{Closure: } u + v \in V,\ \alpha \cdot v \in V$$

Basis — a minimal spanning set; coordinates are the coefficients w.r.t. the basis.

Span — all vectors reachable by linear combinations of a set.

Dimension — number of basis vectors; equals the "degrees of freedom."

Subspace — a vector space living inside another (must contain the zero vector).

1.2 Eigenvalues & eigenvectors

For matrix A, eigenvector v and eigenvalue λ satisfy:

$$\mathrm{A}\cdot\mathrm{v} = \lambda\cdot\mathrm{v}$$

The matrix only stretches or flips v — it doesn't rotate it. λ tells you the stretch factor.

Find eigenvalues: det(A - λ I) = 0 $\leftarrow$ characteristic equation

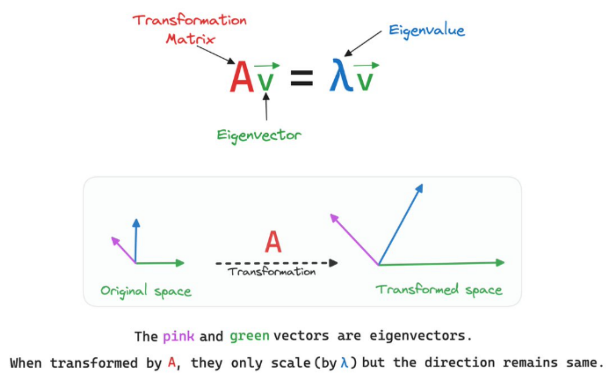

Figure 1: Eigenvectors and Eigenvalues

2 Probability

2.1 Bayes' Theorem

Updating beliefs with evidence:

$$P(H|E) = P(E|H) \cdot P(H) \;/\; P(E)$$

P(H) — prior: your belief about hypothesis before seeing evidence.

P(E|H) — likelihood: how probable the evidence is if H is true.

P(H|E) — posterior: your updated belief after seeing E.

P(E) — marginal: normalising constant (total probability of evidence).

Posterior $\propto$ Likelihood $\times$ Prior. The denominator P(E) just ensures everything sums to 1.

2.2 Gaussian distribution

$$p(x) = \left(\frac{1}{\sigma\sqrt{2\pi}}\right) \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

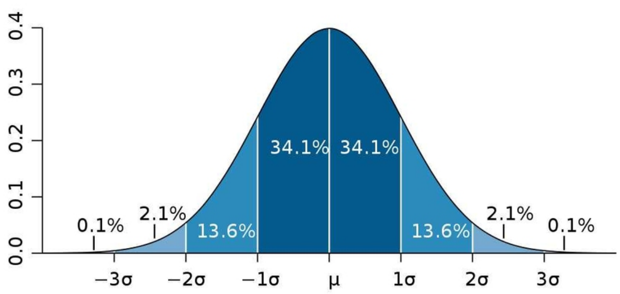

Figure 2: Gaussian Distribution

2.3 KL divergence

How different are two distributions?

KL divergence measures information lost when using distribution Q to approximate distribution P.

$$KL(P \parallel Q) = \sum P(x) \log \frac{P(x)}{Q(x)}$$

Key properties:

KL(P‖Q) $\geq$ 0 always (Gibbs' inequality)

KL(P‖Q) = 0 iff P = Q exactly

Not symmetric: KL(P‖Q) $\neq$ KL(Q‖P) in general

3 Maximum likelihood estimation

The big idea

Given observed data X = x, . . . , x, find parameters that make the data most probable under your model.

$\theta_{MLE} = \arg\max_\theta P(X \mid \theta)$

Since data points are assumed i.i.d., the joint likelihood factors:

$L(\theta) = \prod_i p(x_i \mid \theta)$

We maximise the log-likelihood instead (sum beats product for numerics):

$\ell(\theta) = \sum_i \log p(x_i \mid \theta) \leftarrow$ same argmax, easier math

Step-by-step procedure

1. **Choose a model**
   Specify a parametric distribution for your data, e.g. Gaussian $\mathcal{N}(\mu, \sigma^2)$, Bernoulli $Ber(p)$, or Poisson $Pois(\lambda)$.

2. **Write the log-likelihood**
   Since data points $\{x_1, \ldots, x_n\}$ are assumed i.i.d., the joint likelihood factors, and we maximise:

$$\ell(\theta) = \sum_{i=1}^{n} \log p(x_i \mid \theta)$$

3. **Take the derivative and set to zero**
   Find the critical point by solving:

$$\frac{\partial\, \ell(\theta)}{\partial\, \theta} = 0$$

4. **Solve for θ**
   The solution gives the MLE estimate $\hat{\theta}$.

5. **Verify it is a maximum**
   Check that the second derivative is negative (concave log-likelihood):

$$\left.\frac{\partial^2 \ell(\theta)}{\partial \theta^2}\right|_{\theta=\hat{\theta}} < 0$$

Example: Gaussian MLE. For data drawn from $\mathcal{N}(\mu, \sigma^2)$, the log-likelihood is:

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2$$

Setting derivatives to zero yields:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} x_i \qquad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \hat{\mu})^2$$

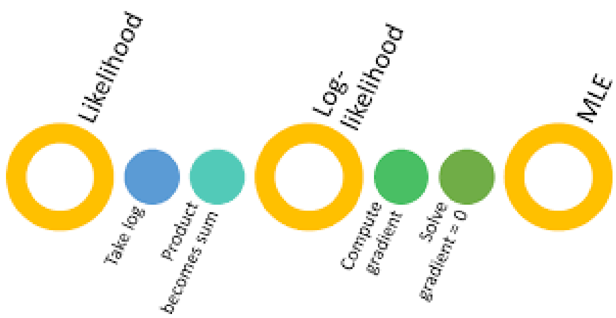

Figure 3: Maximum Likelihood Estimation